Predictive state without privileged hidden states

Hankel/PSR primer · before Notebook 3 · 10–12 minutes plus questions

ILIAD Intensive · Computational Mechanics

What exactly did the probe decode?

HMM belief $b(h)$

A posterior over the hidden states of a **chosen realization**.

Useful and sufficient for that realization's future predictions.

Observable predictive state

Whatever distinctions between histories change the distribution over future observations.

No hidden-state labels required.

What survives if we change the hidden realization but keep every observable word probability fixed?

Histories ask; future tests answer

$$H_{u,v} = \Pr(uv)$$

history u **↓ / future test** v **→**	ϵ	0	1	00	⋯
ϵ	$\Pr(\epsilon)$	$\Pr(0)$	$\Pr(1)$	$\Pr(00)$	⋯
0	$\Pr(0)$	$\Pr(00)$	$\Pr(01)$	$\Pr(000)$	⋯
1	$\Pr(1)$	$\Pr(10)$	$\Pr(11)$	$\Pr(100)$	⋯
⋮	⋮	⋮	⋮	⋮	⋱

Likelihood is not yet prediction

Joint row

$$H_{u,\cdot} = \left(\Pr(uv) \right)_v$$

Its scale includes how likely the history itself was.

Conditional predictive row

$$K_{u,\cdot} = \left(\Pr(v \mid u) \right)_v$$

It compares future profiles after conditioning on the history.

Proportional joint rows can encode identical conditional predictions.

Predictive equivalence quantifies over every future

Histories u and u' are predictively equivalent when

$$\Pr(v \mid u) = \Pr(v \mid u') \qquad \text{for every finite future word } v.$$

One token
A useful probe, but often too weak.

Displayed suffixes
Finite evidence for a pattern.

Every future
The process-relative equivalence notion.

Finite predictive coordinates are earned, not assumed

full future rows → linear dimension → **selected core tests** → symbol updates

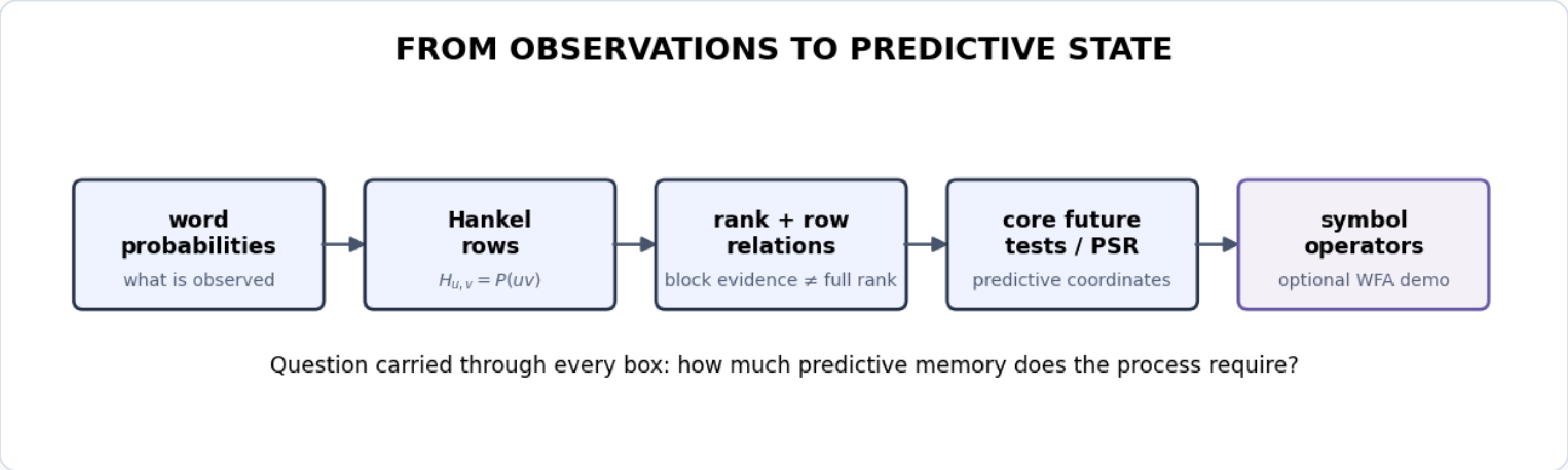
PSR idea

Use probabilities of a small set of future tests as coordinates when those tests span the predictive space.

A finite table is evidence, not yet a theorem about the full process; the notebook earns its later claims before selecting core tests.

Theory anchors: Littman, Sutton & Singh (2001), predictive state representations; Jaeger (2000), observable-operator models and finite linear realizations.

The notebook starts from observations



OPEN NOTEBOOK 3

Interrogate the finite table, distinguish local evidence from a full claim, normalize before comparing predictions, and only then choose coordinates.